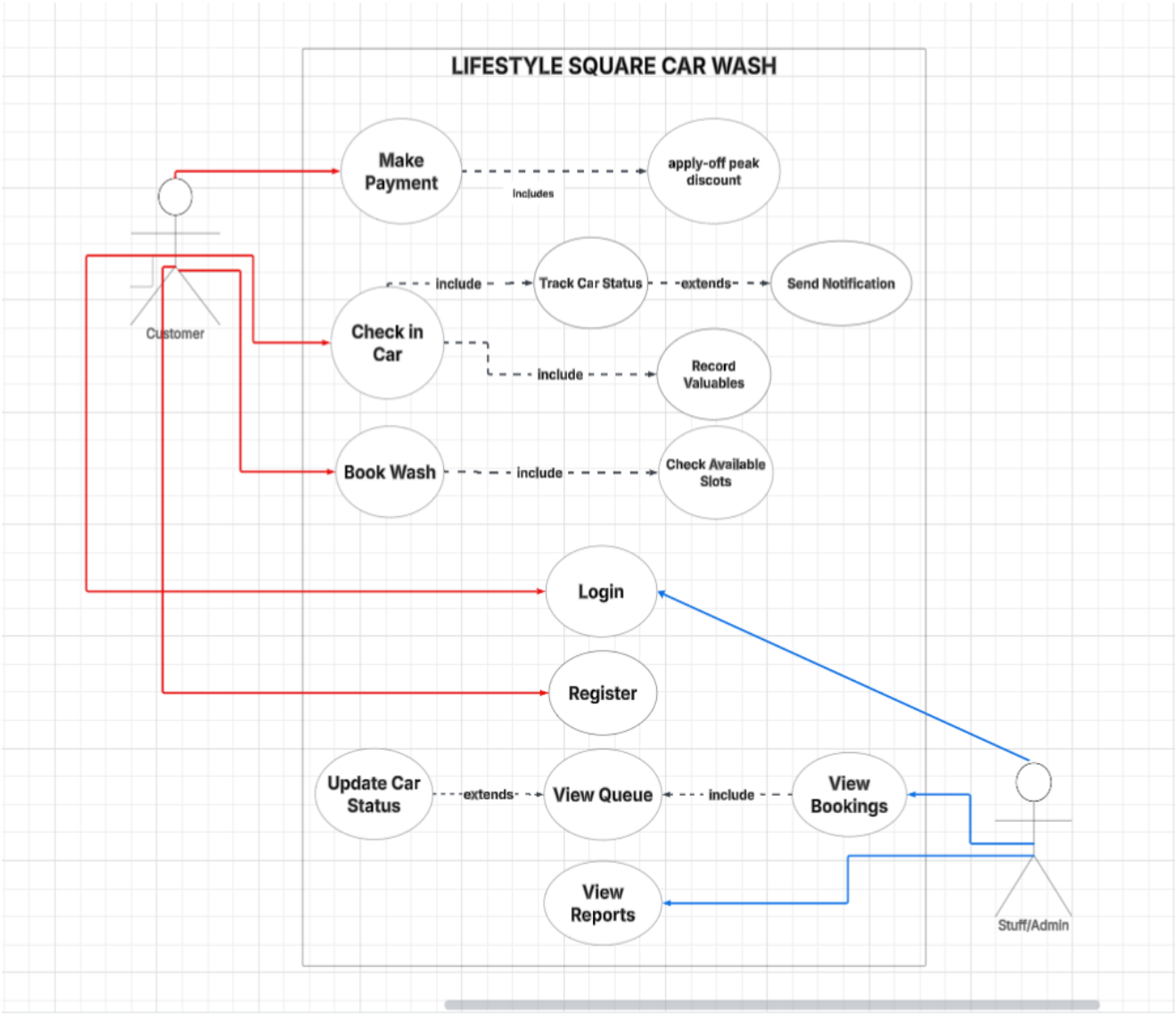
USE CASE DIAGRAM



LUCID WEBSITE LINK: https://lucid.app/lucidchart/f8d4a778-4bb7-4db2-82c1-aa6f739ccbfb/edit?viewport_loc=-1516%2C-1413%2C5464%2C2892%2C0_0&invitationId=inv_0d392055-5ac3-467b-bceb-282571b0644e

Use Case Diagram Explanation: LIFESTYLE SQUARE CAR WASH System

This Use Case Diagram describes the interaction between users and the Lifestyle Square Car Wash System. The system boundary is named LIFESTYLE SQUARE CAR WASH, and it contains all the functionalities.

1. Actors

There are two primary actors outside the system:

- **Customer:** The external user who brings a car for washing services.

- **Staff/Admin:** The internal user who works at the car wash and manages operations.

2. Use Cases for CUSTOMER

The Customer (represented by the red lines) can perform 5 main actions:

- **Register:** Create a new account on the system.

- **Login:** Authenticate and access their account.

- **Book Wash:** Book a slot to wash their car, it includes Check Available Slots -This is a mandatory included process. When booking, the system must check available slots and allow the customer to select from the free slots.

- Check in Car: Physically bring the car to the car wash for the service, this includes Record Valuables - When checking in, customers must record valuables left inside the car for safety and to prevent theft claims also includes Track car status - After check-in, the customer is able to track the progress of their car (e.g., Washing, Drying, Ready).

- Make Payment: Pay for the service after all processes are complete, which includes apply-off peak discount. The system automatically checks the time and applies a discount if the service is booked during off-peak hours.

Additional Relationship:

- Track car status extends send notification: When the status of the car changes, this extended use case triggers, and the system sends an SMS/App notification to the customer.

3. Use Cases for STAFF/ADMIN

The Staff/Admin (represented by the blue lines) can perform 4 main actions:

- Login: Log into the system if they are an existing user.

- **View Bookings:** View all bookings made by customers which includes View Queue -To view bookings effectively, the staff must view the queue of cars waiting to be washed.

- **View Reports:** View system reports such as daily sales, number of washes, and performance reports.

- **Update car status:** Change the status of a car in the system (e.g., from "In Queue" to "Washing" to "Ready for Collection") which extends View Queue This function extends View Queue, meaning the staff updates the status directly from the queue view.

Summary of Flow: A Customer Registers and Logs in, books a Wash (system checks slots), Checks in their Car (valuables are recorded), Staff Updates the car status from the queue, Customer Tracks status and receives notifications, Customer Makes Payment (with possible off-peak discount), and Staff manages the process through View Bookings, View Queue, and View Reports.